Questions 1–7

Complete the notes below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes 1–7 on your answer sheet.

Britain's Industrial Revolution

Steam power

- Newcomen's steam engine was used in mines to remove water.
- In Watt and Boulton's steam engine, the movement of the **1** ..... was linked to a gear system.
- A greater supply of **2** ..... was required to power steam engines.

Textile industry

- Before the Industrial Revolution, spinners and weavers worked at home and in **3** ..... .
- Not as much **4** ..... was needed to produce cloth once the spinning jenny and power loom were invented.

Iron industry

- Smelting of iron ore with coke resulted in material that was better **5** ..... .
- Demand for iron increased with the growth of the **6** ..... .

Communications

- Cooke and Wheatstone patented the first telegraphy system.
- The telegraphy system was used to prevent locomotives colliding.

Urbanisation

- Small towns turned into cities very quickly.
- The new cities were dirty, crowded and lacked sufficient **7** ..... .

Questions 8–13

Do the following statements agree with the information given in Reading Passage 1?

In boxes 8–13 on your answer sheet, write

TRUE	*if the statement agrees with the information*
FALSE	*if the statement contradicts the information*
NOT GIVEN	*if there is no information on this*

8 Britain's canal network grew rapidly so that more goods could be transported around the country.

9 Costs in the iron industry rose when the technique of smelting iron ore with coke was introduced.

10 Samuel Morse's communication system was more reliable than that developed by William Cooke and Charles Wheatstone.

11 The economic benefits of industrialisation were limited to certain sectors of society.

12 Some skilled weavers believed that the introduction of the new textile machines would lead to job losses.

13 There was some sympathy among local people for the Luddites who were arrested near Huddersfield.

→ p. 123

READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 below.*

Athletes and stress

A It isn't easy being a professional athlete. Not only are the physical demands greater than most people could handle, athletes also face intense psychological pressure during competition. This is something that British tennis player Emma Raducanu wrote about on social media following her withdrawal from the 2021 Wimbledon tournament. Though the young player had been doing well in the tournament, she began having difficulty regulating her breathing and heart rate during a match, which she later attributed to 'the accumulation of the excitement and the buzz'.

B For athletes, some level of performance stress is almost unavoidable. But there are many different factors that dictate just how people's minds and bodies respond to stressful events. Typically, stress is the result of an exchange between two factors: demands and resources. An athlete may feel stressed about an event if they feel the demands on them are greater than they can handle. These demands include the high level of physical and mental effort required to succeed, and also the athlete's concerns about the difficulty of the event, their chance of succeeding, and any potential dangers such as injury. Resources, on the other hand, are a person's ability to cope with these demands. These include factors such as the competitor's degree of confidence, how much they believe they can control the situation's outcome, and whether they're looking forward to the event or not.

C Each new demand or change in circumstances affects whether a person responds positively or negatively to stress. Typically, the more resources a person feels they have in handling the situation, the more positive their stress response. This positive stress response is called a challenge state. But should the person feel there are too many demands placed on them, the more likely they are to experience a negative stress response – known as a threat state. Research shows that the challenge states lead to good performance, while threat states lead to poorer performance. So, in Emma Raducanu's case, a much larger audience, higher expectations and facing a more skilful opponent, may all have led her to feel there were greater demands being placed on her at Wimbledon – but she didn't have the resources to tackle them. This led to her experiencing a threat response.

D Our challenge and threat responses essentially influence how our body responds to stressful situations, as both affect the production of adrenaline and cortisol – also known as 'stress hormones'. During a challenge state, adrenaline increases the amount of blood pumped from the heart and expands the blood vessels, which allows more energy to be delivered to the muscles and brain. This increase of blood and decrease of pressure in the blood vessels has been consistently related to superior sport performance in everything from cricket batting, to golf putting and

football penalty taking. But during a threat state, cortisol inhibits the positive effect of adrenaline, resulting in tighter blood vessels, higher blood pressure, slower psychological responses, and a faster heart rate. In short, a threat state makes people more anxious – they make worse decisions and perform more poorly. In tennis players, cortisol has been associated with more unsuccessful serves and greater anxiety.

E That said, anxiety is also a common experience for athletes when they're under pressure. Anxiety can increase heart rate and perspiration, cause heart palpitations, muscle tremors and shortness of breath, as well as headaches, nausea, stomach pain, weakness and a desire to escape in more extreme cases. Anxiety can also reduce concentration and self-control and cause overthinking. The intensity with which a person experiences anxiety depends on the demands and resources they have. Anxiety may also manifest itself in the form of excitement or nervousness depending on the stress response. Negative stress responses can be damaging to both physical and mental health – and repeated episodes of anxiety coupled with negative responses can increase risk of heart disease and depression.

F But there are many ways athletes can ensure they respond positively under pressure. Positive stress responses can be promoted through the language that they and others – such as coaches or parents – use. Psychologists can also help athletes change how they see their physiological responses – such as helping them see a higher heart rate as excitement, rather than nerves. Developing psychological skills, such as visualisation, can also help decrease physiological responses to threat. Visualisation may involve the athlete recreating a mental picture of a time when they performed well, or picturing themselves doing well in the future. This can help create a feeling of control over the stressful event. Recreating competitive pressure during training can also help athletes learn how to deal with stress. An example of this might be scoring athletes against their peers to create a sense of competition. This would increase the demands which players experience compared to a normal training session, while still allowing them to practise coping with stress.

Questions 14–18

Reading Passage 2 has six paragraphs, **A–F**.

Which paragraph contains the following information?

*Write the correct letter, **A–F**, in boxes 14–18 on your answer sheet.*

NB *You may use any letter more than once.*

14 reference to two chemical compounds which impact on performance

15 examples of strategies for minimising the effects of stress

16 how a sportsperson accounted for their own experience of stress

17 study results indicating links between stress responses and performance

18 mention of people who can influence how athletes perceive their stress responses

Questions 19–22

Complete the sentences below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes 19–22 on your answer sheet.

19 Performance stress involves many demands on the athlete, for example, coping with the possible risk of ..... .

20 Cortisol can cause tennis players to produce fewer good ..... .

21 Psychologists can help athletes to view their physiological responses as the effect of a positive feeling such as ..... .

22 ..... is an example of a psychological technique which can reduce an athlete's stress responses.

Questions 23 and 24

*Choose **TWO** letters, **A–E**.*

Write the correct letters in boxes 23 and 24 on your answer sheet.

Which **TWO** facts about Emma Raducanu's withdrawal from the Wimbledon tournament are mentioned in the text?

A the stage at which she dropped out of the tournament
B symptoms of her performance stress at the tournament
C measures which she had taken to manage her stress levels
D aspects of the Wimbledon tournament which increased her stress levels
E reactions to her social media posts about her experience at Wimbledon

Questions 25 and 26

*Choose **TWO** letters, **A–E**.*

Write the correct letters in boxes 25 and 26 on your answer sheet.

Which **TWO** facts about anxiety are mentioned in Paragraph E of the text?

A the factors which determine how severe it may be
B how long it takes for its effects to become apparent
C which of its symptoms is most frequently encountered
D the types of athletes who are most likely to suffer from it
E the harm that can result if athletes experience it too often

→ p. 123

READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27–40**, which are based on Reading Passage 3 below.*

An inquiry into the existence of the gifted child

Let us start by looking at a modern 'genius', Maryam Mirzakhani, who died at the early age of 40. She was the only woman to win the Fields Medal – the mathematical equivalent of a Nobel prize. It would be easy to assume that someone as special as Mirzakhani must have been one of those 'gifted' children, those who have an extraordinary ability in a specific sphere of activity or knowledge. But look closer and a different story emerges. Mirzakhani was born in Tehran, Iran. She went to a highly selective girls' school but maths wasn't her interest – reading was. She loved novels and would read anything she could lay her hands on. As for maths, she did rather poorly at it for the first couple of years in her middle school, but became interested when her elder brother told her about what he'd learned. He shared a famous maths problem from a magazine that fascinated her – and she was hooked.

In adult life it is clear that she was curious, excited by what she did and also resolute in the face of setbacks. One of her comments sums it up. 'Of course, the most rewarding part is the "Aha" moment, the excitement of discovery and enjoyment of understanding something new . . . But most of the time, doing mathematics for me is like being on a long hike with no trail and no end in sight.' That trail took her to the heights of original research into mathematics.

Is her background unusual? Apparently not. Most Nobel prize winners were unexceptional in childhood. Einstein was slow to talk as a baby. He failed the general part of the entry test to Zurich Polytechnic – though they let him in because of high physics and maths scores. He struggled at work initially, but he kept plugging away and eventually rewrote the laws of Newtonian mechanics with his theory of relativity.

There has been a considerable amount of research on high performance over the last century that suggests it goes way beyond tested intelligence. On top of that, research is clear that brains are flexible, new neural pathways can be created, and IQ isn't fixed. For example, just because you can read stories with hundreds of pages at the age of five doesn't mean you will still be ahead of your contemporaries in your teens.

While the jury is out on giftedness being innate and other factors potentially making the difference, what is certain is that the behaviours associated with high levels of performance are replicable and most can be taught – even traits such as curiosity.

According to my colleague Prof Deborah Eyre, with whom I've collaborated on the book *Great Minds and How to Grow Them*, the latest neuroscience and psychological research suggests most individuals can reach levels of performance associated in school with the gifted and talented. However, they must be taught the right attitudes and approaches to their learning and develop the attributes of

high performers – curiosity, persistence and hard work, for example – an approach Eyre calls 'high performance learning'. Critically, they need the right support in developing those approaches at home as well as at school.

Prof Anders Ericsson, an eminent education psychologist at Florida State University, US, is the co-author of *Peak: Secrets from the New Science of Expertise*. After research going back to 1980 into diverse achievements, from music to memory to sport, he doesn't think unique and innate talents are at the heart of performance. Deliberate practice, that stretches you every step of the way, and around 10,000 hours of it, is what produces the goods. It's not a magic number – the highest performers move on to doing a whole lot more, of course. Ericsson's memory research is particularly interesting because random students, trained in memory techniques for the study, went on to outperform others thought to have innately superior memories – those who you might call gifted.

But it is perhaps the work of Benjamin Bloom, another distinguished American educationist working in the 1980s, that gives the most pause for thought. Bloom's team looked at a group of extraordinarily high achieving people in disciplines as varied as ballet, swimming, piano, tennis, maths, sculpture and neurology. He found a pattern of parents encouraging and supporting their children, often in areas they enjoyed themselves. Bloom's outstanding people had worked very hard and consistently at something they had become hooked on when at a young age, and their parents all emerged as having strong work ethics themselves.

Eyre says we know how high performers learn. From that she has developed a high performing learning approach. She is working on this with a group of schools, both in Britain and abroad. Some spin-off research, which looked in detail at 24 of the 3,000 children being studied who were succeeding despite difficult circumstances, found something remarkable. Half were getting free school meals because of poverty, more than half were living with a single parent, and four in five were living in disadvantaged areas. Interviews uncovered strong evidence of an adult or adults in the child's life who valued and supported education, either in the immediate or extended family or in the child's wider community. Children talked about the need to work hard at school, to listen in class and keep trying.

Let us end with Einstein, the epitome of a genius. He clearly had curiosity, character and determination. He struggled against rejection in early life but was undeterred. Did he think he was a genius or even gifted? He once wrote: 'It's not that I'm so smart, it's just that I stay with problems longer. Most people say it is the intellect which makes a great scientist. They are wrong: it is character.'

Questions 27–32

*Complete the summary using the list of phrases, **A–K**, below.*

*Write the correct letter, **A–K**, in boxes 27–32 on your answer sheet.*

Maryam Mirzakhani

Maryam Mirzakhani is regarded as **27** ..... in the field of mathematics because she was the only female holder of the prestigious Fields Medal – a record that she retained at the time of her death. However, maths held little **28** ..... for her as a child and in fact her performance was below average until she was **29** ..... by a difficult puzzle that one of her siblings showed her.

Later, as a professional mathematician, she had an inquiring mind and proved herself to be **30** ..... when things did not go smoothly. She said she got the greatest **31** ..... from making ground-breaking discoveries and in fact she was responsible for some extremely **32** ..... mathematical studies.

A appeal	**B** determined	**C** intrigued
D single	**E** achievement	**F** devoted
G involved	**H** unique	**I** innovative
J satisfaction	**K** intent	

Questions 33–37

Do the following statements agree with the claims of the writer in Reading Passage 3?

In boxes 33–37 on your answer sheet, write

YES	*if the statement agrees with the claims of the writer*
NO	*if the statement contradicts the claims of the writer*
NOT GIVEN	*if it is impossible to say what the writer thinks about this*

33 Many people who ended up winning prestigious intellectual prizes only reached an average standard when young.

34 Einstein's failures as a young man were due to his lack of confidence.

35 It is difficult to reach agreement on whether some children are actually born gifted.

36 Einstein was upset by the public's view of his life's work.

37 Einstein put his success down to the speed at which he dealt with scientific questions.

Questions 38–40

*Choose the correct letter, **A**, **B**, **C** or **D**.*

Write the correct letter in boxes 38–40 on your answer sheet.

38 What does Eyre believe is needed for children to equal 'gifted' standards?

A strict discipline from the teaching staff
B assistance from their peers in the classroom
C the development of a spirit of inquiry towards their studies
D the determination to surpass everyone else's achievements

39 What is the result of Ericsson's research?

A Very gifted students do not need to work on improving memory skills.
B Being born with a special gift is not the key factor in becoming expert.
C Including time for physical exercise is crucial in raising performance.
D 10,000 hours of relevant and demanding work will create a genius.

40 In the penultimate paragraph, it is stated the key to some deprived children's success is

A a regular and nourishing diet at home.
B the loving support of more than one parent.
C a community which has well-funded facilities for learning.
D the guidance of someone who recognises the benefits of learning.

→ p. 123

WRITING

WRITING TASK 1

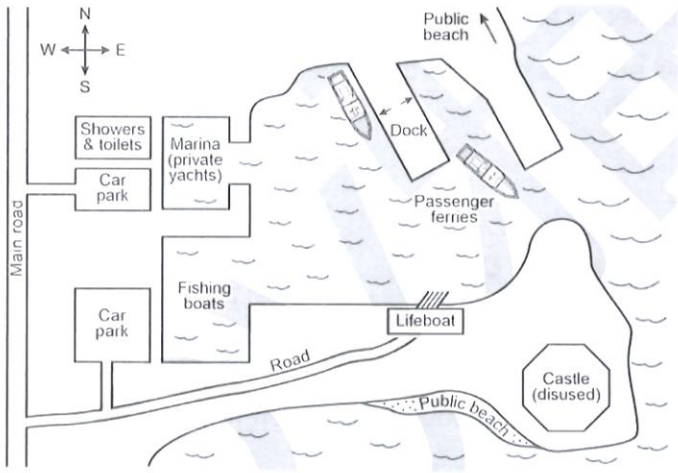
You should spend about 20 minutes on this task.

The plans below show a harbour in 2000 and how it looks today.

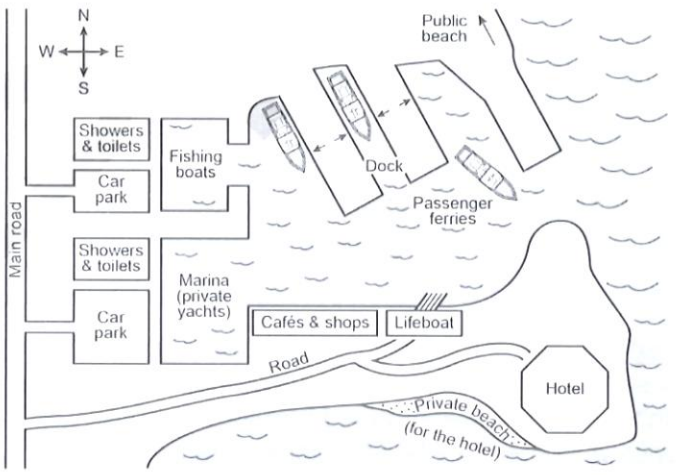
Summarise the information by selecting and reporting the main features, and make comparisons where relevant.

Write at least 150 words.

Porth Harbour (in 2000)



Porth Harbour (today)



 → p. 130

WRITING TASK 2

You should spend about 40 minutes on this task.

Write about the following topic:

> ***The working week should be shorter and workers should have a longer weekend.***
>
> ***Do you agree or disagree?***

Give reasons for your answer and include any relevant examples from your own knowledge or experience.

Write at least 250 words.

→ p. 131

SPEAKING

PART 1

The examiner asks you about yourself, your home, work or studies and other familiar topics.

EXAMPLE

Travelling by plane

- Have you travelled a lot by plane? [To where?/Why not?]
- Why do you think some people enjoy travelling by plane?
- Would you like to live near an airport? [Why/Why not?]
- In the future, do you think that you will travel by plane more often? [Why/Why not?]

PART 2

Describe a person from your country who has won a prize, award or medal.

You should say:
who this person is
which prize, award or medal they received
what they did to win this

and explain whether you think it was right that this person received this prize, award or medal.

You will have to talk about the topic for one to two minutes. You have one minute to think about what you are going to say. You can make some notes to help you if you wish.

PART 3

Discussion topics:

Rewards for children at school

Example questions:
What types of school prizes do children in your country receive?
What do you think are the advantages of rewarding schoolchildren for good work?
Do you agree that it's more important for children to receive rewards from their parents than from teachers?

Rewards for sportspeople

Example questions:
Do you think that some sportspeople (e.g., top footballers) are paid too much money?
Should everyone on a team get the same prize money when they win?
Do you agree with the view that, in sport, taking part is more important than winning?

Test 3

LISTENING

PART 1 *Questions 1–10*

Questions 1–6

Complete the notes below.

*Write **ONE WORD AND/OR A NUMBER** for each answer.*

Local food shops

Where to go

- Kite Place – near the **1** .....

Fish market

- cross the **2** ..... and turn right
- best to go before **3** ..... pm, earlier than closing time

Organic shop

- called **4** '.....'
- below a restaurant in the large, grey building
- look for the large **5** ..... outside

Supermarket

- take a **6** ..... minibus, number 289

Questions 7–10

Complete the table below.

*Write **ONE WORD ONLY** for each answer.*

Shopping		
	To buy	**Other ideas**
Fish market	a dozen prawns	a handful of **7** ..... (type of seaweed)
Organic shop	beans and a **8** ..... for dessert	spices and **9** .....
Bakery	a brown loaf	a **10** ..... tart

PART 2 *Questions 11–20*

Questions 11–16

What information is given about each of the following festival workshops?

*Choose **SIX** answers from the box and write the correct letter, **A–H**, next to Questions 11–16.*

	Information
A	involves painting and drawing
B	will be led by a prize-winning author
C	is aimed at children with a disability
D	involves a drama activity
E	focuses on new relationships
F	is aimed at a specific age group
G	explores an unhappy feeling
H	raises awareness of a particular culture

Festival workshops

11 Superheroes .....

12 Just do it .....

13 Count on me .....

14 Speak up .....

15 Jump for joy .....

16 Sticks and stones .....

Questions 17 and 18

*Choose **TWO** letters, **A–E**.*

Which **TWO** reasons does the speaker give for recommending *Alive and Kicking*?

A It will appeal to both boys and girls.
B The author is well known.
C It has colourful illustrations.
D It is funny.
E It deals with an important topic.

Questions 19 and 20

*Choose **TWO** letters, **A–E**.*

Which **TWO** pieces of advice does the speaker give to parents about reading?

A Encourage children to write down new vocabulary.
B Allow children to listen to audio books.
C Get recommendations from librarians.
D Give children a choice about what they read.
E Only read aloud to children until they can read independently.

PART 3 *Questions 21–30*

Questions 21–25

*Choose the correct letter, **A**, **B** or **C**.*

Science experiment for Year 12 students

21 How does Clare feel about the students in her Year 12 science class?

A worried that they are not making progress
B challenged by their poor behaviour in class
C frustrated at their lack of interest in the subject

22 How does Jake react to Clare's suggestion about an experiment based on children's diet?

A He is concerned that the results might not be meaningful.
B He feels some of the data might be difficult to obtain.
C He suspects that the conclusions might be upsetting.

23 What problem do they agree may be involved in an experiment involving animals?

A Any results may not apply to humans.
B It may be complicated to get permission.
C Students may not be happy about animal experiments.

24 What question do they decide the experiment should address?

A Are mice capable of controlling their food intake?
B Does an increase in sugar lead to health problems?
C How much do supplements of different kinds affect health?

25 Clare might also consider doing another experiment involving

A other types of food supplement.
B different genetic strains of mice.
C varying amounts of exercise.

Questions 26–30

Complete the flowchart below.

*Choose **FIVE** answers from the box and write the correct letter, **A–H**, next to Questions 26–30.*

A	size
B	escape
C	age
D	water
E	cereal
F	calculations
G	changes
H	colour

Choose mice which are all the same **26** ..... .

↓

Divide the mice into two groups, each with a different **27** ..... .

↓

Put each group in a separate cage.
Feed group A commercial mouse food.
Feed group B the same, but also sugar contained in **28** ..... .

↓

Take measurements using an electronic scale.
Place them in a weighing chamber to prevent **29** ..... .

↓

Do all necessary **30** ..... .

PART 4 *Questions 31–40*

Complete the notes below.

*Write **ONE WORD ONLY** for each answer.*

Microplastics

Where microplastics come from

- fibres from some **31** ..... during washing
- the breakdown of large pieces of plastic
- waste from industry
- the action of vehicle tyres on roads

Effects of microplastics

- They cause injuries to the **32** ..... of wildlife and affect their digestive systems.
- They enter the food chain, e.g., in bottled and tap water, **33** ..... and seafood.
- They may not affect human health, but they are already banned in skin cleaning products and **34** ..... in some countries.
- Microplastics enter the soil through the air, rain and **35** ..... .

Microplastics in the soil – a study by Anglia Ruskin University

- Earthworms are important because they add **36** ..... to the soil.
- The study aimed to find whether microplastics in earthworms affect the **37** ..... of plants.
- The study found that microplastics caused:
  - **38** ..... loss in earthworms
  - fewer seeds to germinate
  - a rise in the level of **39** ..... in the soil.

The study concluded:

- soil should be seen as an important natural process.
- changes to soil damage both ecosystems and **40** ..... .

→ p. 124 | p. 112

READING

READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.*

Archaeologists discover evidence of prehistoric island settlers

In early April 2019, Dr Ceri Shipton and his colleagues from Australian National University became the first archaeologists to explore Obi, one of many tropical islands in Indonesia's Maluku Utara province. The research team's discoveries suggest that the prehistoric people who lived on Obi were adept on both land and sea, hunting in the dense rainforest, foraging on the seashore, and possibly even voyaging between islands.

The excavations were part of a project to learn more about how people first dispersed from mainland Asia, through the Indonesian archipelago and into the prehistoric continent that once connected Australia and New Guinea. The team's earlier research suggested that the northernmost islands in the group, known as the Wallacean islands, including Obi, would have offered the easiest migration route. It also seemed likely that these islands were crucial 'stepping stones' on humans' island-hopping voyages through this region millennia ago. But to support this idea, they needed archaeological evidence for humans living in this remote area in the ancient past. So, they travelled to Obi to look for sites that might reveal evidence of early occupation.

Just inland from the village of Kelo on Obi's northern coast, Shipton and his colleagues found two caves containing prehistoric rock shelters that were suitable for excavation. With the permission and help of the local people of Kelo, they dug a small test excavation in each shelter. There they found numerous artefacts, including fragments of axes, some dating to about 14,000 years ago. The earliest axes at Kelo were made using clam shells. Axes made from clam shells from roughly the same time had also previously been found elsewhere in this region, including on the nearby island of Gebe to the northeast. As on Gebe, it is highly likely that Obi's axes were used in the construction of canoes, thus allowing these early peoples to maintain connections between communities on neighbouring islands.

The oldest cultural layers from the Kelo site provided the team with the earliest record for human occupation on Obi, dating back around 18,000 years. At this time the climate was drier and colder than today, and the island's dense rainforests would likely have been much less impenetrable than they are now. Sea levels were about 120 metres lower, meaning Obi was a much larger island, encompassing what is today the separate island of Bisa, as well as several other small islands nearby.

Roughly 11,700 years ago, as the most recent ice age ended, the climate became significantly warmer and wetter, no doubt making Obi's jungle much thicker. According to the researchers, it is no coincidence that around this time the first axes crafted from stone rather than sea shells appear, likely in response to their heavy-duty use for clearing and modification of the increasingly dense rainforest. While stone takes about twice as long to grind into an axe compared to shell, the harder material keeps its sharp edge for longer.

Judging by the bones which the researchers unearthed in the Kelo caves, people living there mainly hunted the Rothschild's cuscus, a possum-like creature that still lives on Obi today. As the forest grew more dense, people probably used axes to clear patches of forest and make hunting easier.

Shipton's team's excavation of the shelters at the Kelo site unearthed a volcanic glass substance called obsidian, which must have been brought over from another island, as there is no known source on Obi. It also revealed particular types of beads, similar to those previously found on islands in southern Wallacea. These finds again support the idea that Obi islanders routinely travelled to other islands.

The excavations suggest people successfully lived in the two Kelo shelters for about 10,000 years. But then, about 8,000 years ago, both were abandoned. Did the residents leave Obi completely, or move elsewhere on the island? Perhaps the jungle had grown so thick that axes were no longer a match for the dense undergrowth. Perhaps people simply moved to the coast and turned to fishing rather than hunting as a means of survival.

Whatever the reason for the departure, there is no evidence for use of the Kelo shelters after this time, until about 1,000 years ago, when they were re-occupied by people who owned pottery as well as items made out of gold and silver. It seems likely, in view of Obi's location, that this final phase of occupation also saw the Kelo shelters used by people involved in the historic trade in spices between the Maluku islands and the rest of the world.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1–7 on your answer sheet, write

TRUE	*if the statement agrees with the information*
FALSE	*if the statement contradicts the information*
NOT GIVEN	*if there is no information on this*

1 Archaeological research had taken place on the island of Obi before the arrival of Ceri Shipton and his colleagues.

2 At the Kelo sites, the researchers found the first clam shell axes ever to be discovered in the region.

3 The size of Obi today is less than it was 18,000 years ago.

4 A change in the climate around 11,700 years ago had a greater impact on Obi than on the surrounding islands.

5 The researchers believe there is a connection between warmer, wetter weather and a change in the material used to make axes.

6 Shipton's team were surprised to find evidence of the Obi islanders' hunting practices.

7 It is thought that the Kelo shelters were occupied continuously until about 1,000 years ago.

Questions 8–13

Complete the notes below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes 8–13 on your answer sheet.

Archaeological findings on Obi

Excavations of rock shelters inside **8** ..... near the village of Kelo revealed:

- axes from around 14,000 years ago, probably used to make canoes
- axes made out of **9** ..... , dating from around 11,700 years ago
- **10** ..... of an animal: evidence of what ancient islanders ate
- evidence of travel between islands:
  - obsidian: a material that is not found naturally on Obi
  - **11** ..... which resembled ones found on other islands.

It is thought that from 8,000 years ago, Obi islanders:

- may have switched from hunting to fishing
- had **12** ..... as well as items made out of metal
- probably took part in the production and sale of **13** ..... .

→ p. 125

READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 below.*

The global importance of wetlands

A Wetlands are areas where water covers the soil, or is present either at or near the surface of the soil, for all or part of the year. These are complex ecosystems, rich in unique plant and animal life. But according to the World Wide Fund for Nature (WWFN), half of the world's wetlands have disappeared since 1990 – converted or destroyed for commercial development, drainage schemes and the extraction of minerals and peat*. Many of those that remain have been damaged by agricultural pesticides and fertilisers, industrial pollutants, and construction works.

B Throughout history, humans have gathered around wetlands, and their fertile ecosystems have played an important part in human development. Consequently, they are of considerable religious, historical and archaeological value to many communities around the world. 'Wetlands directly support the livelihoods and well-being of millions of people,' says Dr Matthew McCartney, principal researcher and hydrologist at the International Water Management Institute (IWMI). 'In many developing countries, large numbers of people are dependent on wetland agriculture for their livelihoods.'

C They also serve a crucial environmental purpose. 'Wetlands are one of the key tools in mitigating climate change across the planet,' says Pieter van Eijk, head of Climate Adaptation at Wetlands International (WI), pointing to their use as buffers that protect coastal areas from sea-level rise and extreme weather events such as hurricanes and flooding. Wetland coastal forests provide food and water, as well as shelter from storms, and WI and other agencies are working to restore those forests which have been lost. 'It can be as simple as planting a few trees per hectare to create shade and substantially change a microclimate,' he says. 'Implementing climate change projects isn't so much about money.'

D The world's wetlands are, unfortunately, rich sources for in-demand commodities, such as palm oil and pulpwood. Peatlands – wetlands with a waterlogged organic soil layer – are particularly targeted. When peatlands are drained for cultivation, they become net carbon emitters instead of active carbon stores, and, according to Marcel Silvius, head of Climate-smart Land-use at WI, this practice causes six per cent of all global carbon emissions. The clearance of peatlands for planting also increases the risk of forest fires, which release huge amounts of CO_2. 'We're seeing huge peatland forests with extremely high biodiversity value being lost for a few decades of oil palm revenues,' says Silvius.

* peat: a brown deposit formed by the partial decomposition of vegetation in wet acidic conditions, often cut out and dried for use as fuel

E The damage starts when logging companies arrive to clear the trees. They dig ditches to enter the peat swamps by boat and then float the logs out the same way. These are then used to drain water out of the peatlands to allow for the planting of corn, oil palms or pulpwood trees. Once the water has drained away, bacteria and fungi then break down the carbon in the peat and turn it into CO_2 and methane. Meanwhile, the remainder of the solid matter in the peat starts to move downwards, in a process known as subsidence**. Peat comprises 90 per cent water, so this is one of the most alarming consequences of peatland clearances. 'In the tropics, peat subsides at about four centimetres a year, so within half a century, very large landscapes on Sumatra and Borneo will become flooded as the peat drops below water level,' says Silvius. 'It's a huge catastrophe that's in preparation. Some provinces will lose 40 per cent of their landmass.'

F And while these industries affect wetlands in ways that can easily be documented, Dr Dave Tickner of the WWFN believes that more subtle impacts can be even more devastating. 'Sediment run-off and fertilisers can be pretty invisible,' says Tickner. 'Over-extraction of water is equally invisible. You do get shock stories about rivers running red, or even catching fire, but there's seldom one big impact that really hurts a wetland.' Tickner does not blame anyone for deliberate damage, however. 'I've worked on wetland issues for 20 years and have never met anybody who wanted to damage a wetland,' he says. 'It isn't something that people generally set out to do. Quite often, the effects simply come from people trying to make a living.'

G Silvius also acknowledges the importance of income generation. 'It's not that we just want to restore the biodiversity of wetlands – which we do – but we recognise there's a need to provide an income for local people.' This approach is supported by IWMI. 'The idea is that people in a developing country will only protect wetlands if they value and profit from them,' says McCartney. 'For sustainability, it's essential that local people are involved in wetland planning and decision making and have clear rights to use wetlands.'

H The fortunes of wetlands would be improved, Silvius suggests, if more governments recognised their long-term value. 'Different governments have different attitudes,' he says, and goes on to explain that some countries place a high priority on restoring wetlands, while others still deny the issue. McCartney is cautiously optimistic, however. 'Awareness of the importance of wetlands is growing,' he says. 'It's true that wetland degradation still continues at a rapid pace, but my impression is that things are slowly changing.'

** subsidence: the sinking of the Earth's surface, most often caused by the removal of resources from the ground

Questions 14–17

Reading Passage 2 has eight paragraphs, **A–H**.

Which paragraph contains the following information?

*Write the correct letter, **A–H**, in boxes 14–17 on your answer sheet.*

14 reference to the need to ensure that inhabitants of wetland regions continue to benefit from them

15 the proportion of wetlands which have already been lost

16 reference to the idea that people are beginning to appreciate the value of wetlands

17 mention of the cultural significance of wetlands

Questions 18–22

Complete the sentences below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes 18–22 on your answer sheet.

18 Peatlands which have been drained begin to release ..... instead of storing it.

19 Once peatland areas have been cleared, ... are more likely to occur.

20 Clearing peatland forests to make way for oil palm plantations destroys the ..... of the local environment.

21 Water is drained out of peatlands through the ..... which are created by logging companies.

22 Draining peatlands leads to ..... : a serious problem which can eventually result in coastal flooding and land loss.

Questions 23–26

Look at the following statements (Questions 23–26) and the list of experts below.

Match each statement with the correct expert, ***A–D****.*

Write the correct letter, ***A–D****, in boxes 23–26 on your answer sheet.*

23 Communities living in wetland regions must be included in discussions about the future of these areas.

24 Official policies towards wetlands vary from one nation to the next.

25 People cause harm to wetlands without having any intention to do so.

26 Initiatives to reverse environmental damage need not be complex.

List of Experts

A Matthew McCartney

B Pieter van Eijk

C Marcel Silvius

D Dave Tickner

.

→ p. 125

READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27–40**, which are based on Reading Passage 3 below.*

Is the era of artificial speech translation upon us?

Once the stuff of science fiction, technology that enables people to talk using different languages is now here. But how effective is it?

Noise, Alex Waibel tells me, is one of the major challenges that artificial speech translation has to meet. A device may be able to recognise speech in a laboratory, or a meeting room, but will struggle to cope with the kind of background noise I can hear in my office surrounding Professor Waibel as he speaks to me from Kyoto station in Japan. I'm struggling to follow him in English, on a scratchy line that reminds me we are nearly 10,000 kilometres apart – and that distance is still an obstacle to communication even if you're speaking the same language, as we are. We haven't reached the future yet. If we had, Waibel would have been able to speak more comfortably in his native German and I would have been able to hear his words in English.

At Karlsruhe Institute of Technology, where he is a professor of computer science, Waibel and his colleagues already give lectures in German that their students can follow in English via an electronic translator. The system generates text that students can read on their laptops or phones, so the process is somewhat similar to subtitling. It helps that lecturers speak clearly, don't have to compete with background chatter, and say much the same thing each year.

The idea of artificial speech translation has been around for a long time. Douglas Adams' science fiction novel, *The Hitchhiker's Guide to the Galaxy*, published in 1979, featured a life form called the 'Babel fish' which, when placed in the ear, enabled a listener to understand any language in the universe. It came to represent one of those devices that technology enthusiasts dream of long before they become practically realisable, like TVs flat enough to hang on walls: objects that we once could only dream of having but that are now commonplace. Now devices that look like prototype Babel fish have started to appear, riding a wave of advances in artificial translation and voice recognition.

At this stage, however, they seem to be regarded as eye-catching novelties rather than steps towards what Waibel calls 'making a language-transparent society'. They tend to be domestic devices or applications suitable for hotel check-ins, for example, providing a practical alternative to speaking traveller's English. The efficiency of the translator is less important than the social function. However, 'Professionals are less inclined to be patient in a conversation,' founder and CEO at Waverly Labs, Andrew Ochoa, observes. To redress this, Waverly is now preparing a new model for professional applications, which entails performance improvements in speech recognition, translation accuracy and the time it takes to deliver the translated speech.

For a conversation, both speakers need to have devices called Pilots (translator earpieces) in their ears. 'We find that there's a barrier with sharing one of the earphones with a stranger,' says Ochoa. That can't have been totally unexpected. The problem would be solved if earpiece translators became sufficiently prevalent that strangers would be likely to already have their own in their ears. Whether that happens, and how quickly, will probably depend not so much on the earpieces themselves, but on the prevalence of voice-controlled devices and artificial translation in general.

Waibel highlights the significance of certain Asian nations, noting that voice translation has really taken off in countries such as Japan with a range of systems. There is still a long way to go, though. A translation system needs to be simultaneous, like the translator's voice speaking over the foreign politician being interviewed on the TV, rather than in sections that oblige speakers to pause after every few remarks and wait for the translation to be delivered. It needs to work offline, for situations where internet access isn't possible, and to address apprehensions about the amount of private speech data accumulating in the cloud, having been sent to servers for processing.

Systems not only need to cope with physical challenges such as noise, they will also need to be socially aware by addressing people in the right way. Some cultural traditions demand solemn respect for academic status, for example, and it is only polite to respect this. Etiquette-sensitive artificial translators could relieve people of the need to know these differing cultural norms. At the same time, they might help to preserve local customs, slowing the spread of habits associated with international English, such as its readiness to get on first-name terms.

Professors and other professionals will not outsource language awareness to software, though. If the technology matures into seamless, ubiquitous artificial speech translation, it will actually add value to language skills. Whether it will help people conduct their family lives or relationships is open to question – though one noteworthy possibility is that it could overcome the language barriers that often arise between generations after migration, leaving children and their grandparents without a shared language.

Whatever uses it is put to, though, it will never be as good as the real thing. Even if voice-morphing technology simulates the speaker's voice, their lip movements won't match, and they will look like they are in a dubbed movie. The contrast will underline the value of shared languages, and the value of learning them. Sharing a language can promote a sense of belonging and community, as with the international scientists who use English as a lingua franca, where their predecessors used Latin. Though the practical need for a common language will diminish, the social value of sharing one will persist. And software will never be a substitute for the subtle but vital understanding that comes with knowledge of a language.

Questions 27–30

*Choose the correct letter, **A**, **B**, **C** or **D**.*

Write the correct letter in boxes 27–30 on your answer sheet.

27 What does the reader learn about the conversation in the first paragraph?

A The speakers are communicating in different languages.
B Neither of the speakers is familiar with their environment.
C The topic of the conversation is difficult for both speakers.
D Aspects of the conversation are challenging for both speakers.

28 What assists the electronic translator during lectures at Karlsruhe Institute of Technology?

A the repeated content of lectures
B the students' reading skills
C the languages used
D the lecturers' technical ability

29 When referring to *The Hitchhiker's Guide to the Galaxy,* the writer suggests that

A the Babel fish was considered undesirable at the time.
B this book was not seriously intending to predict the future.
C artificial speech translation was not a surprising development.
D some speech translation techniques are better than others.

30 What does the writer say about sharing earpieces?

A It is something people will get used to doing.
B The reluctance to do this is understandable.
C The equipment will be unnecessary in the future.
D It is something few people need to worry about.

Questions 31–34

Complete each sentence with the correct ending, ***A–F,*** *below.*

Write the correct letter, ***A–F,*** *in boxes 31–34 on your answer sheet.*

31 Speech translation methods are developing fast in Japan

32 TV interviews that use translation voiceover methods are successful

33 Future translation systems should address people appropriately

34 Users may be able to maintain their local customs

A	but there are concerns about this.
B	as systems do not need to conform to standard practices.
C	but they are far from perfect.
D	despite the noise issues.
E	because translation is immediate.
F	and have an awareness of good manners.

Questions 35–40

Do the following statements agree with the views of the writer in Reading Passage 3?

In boxes 35–40 on your answer sheet, write

YES	*if the statement agrees with the views of the writer*
NO	*if the statement contradicts the views of the writer*
NOT GIVEN	*if it is impossible to say what the writer thinks about this*

35 Language translation systems will be seen as very useful throughout the academic and professional worlds.

36 The overall value of automated translation to family life is yet to be shown.

37 Automated translation could make life more difficult for immigrant families.

38 Visual aspects of language translation are being considered by scientists.

39 International scientists have found English easier to translate into other languages than Latin.

40 As far as language is concerned, there is a difference between people's social and practical needs.

→ p. 125

WRITING

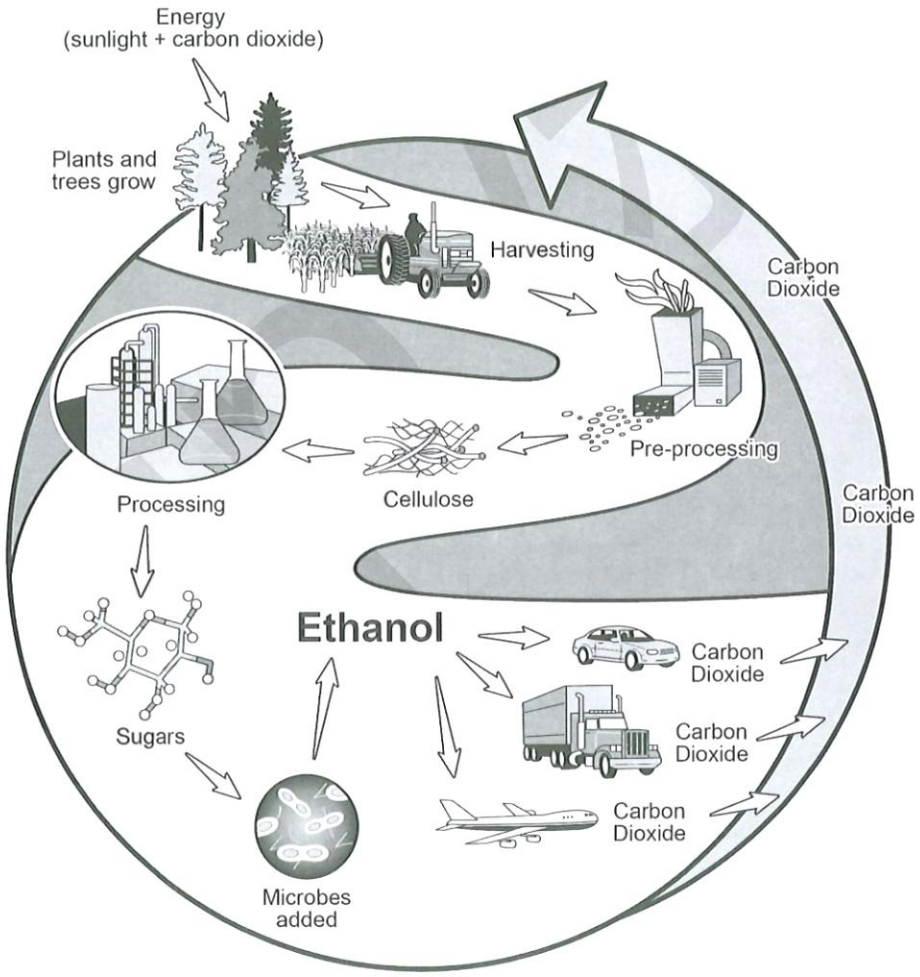
WRITING TASK 1

You should spend about 20 minutes on this task.

> ***The diagram below shows how a biofuel called ethanol is produced.***
>
> ***Summarise the information by selecting and reporting the main features, and make comparisons where relevant.***

Write at least 150 words.

Biofuel production: how ethanol is made



→ p. 133

WRITING TASK 2

You should spend about 40 minutes on this task.

Write about the following topic:

> ***It is important for everyone, including young people, to save money for their future.***
>
> ***To what extent do you agree or disagree with this statement?***

Give reasons for your answer and include any relevant examples from your own knowledge or experience.

Write at least 250 words.

→ p. 134

SPEAKING

PART 1

The examiner asks you about yourself, your home, work or studies and other familiar topics.

EXAMPLE

Holidays

- Do you prefer spending holidays with friends or with family? [Why?]
- What kind of holiday accommodation do you like to stay in? [Why?]
- What plans do you have for your next holiday?
- Is your city or region a good place for other people to visit on holiday? [Why/Why not?]

PART 2

Describe a car journey you made that took longer than expected.

You should say:
where you were going
who you were with
how you felt during the journey

and explain why this car journey took longer than expected.

You will have to talk about the topic for one to two minutes. You have one minute to think about what you are going to say. You can make some notes to help you if you wish.

PART 3

Discussion topics:

Driving a car

Example questions:
How interested are young people in your country in learning to drive?
What are the differences between driving in the countryside and driving in the city?
Do you consider most drivers where you live to be good drivers?

Electric cars

Example questions:
How popular are electric cars in your country?
In what ways could more people be persuaded to buy electric cars?
Do you think all cars will be electric one day?

Test 4

LISTENING

PART 1 *Questions 1–10*

Questions 1–6

Complete the notes below.

*Write **ONE WORD AND/OR A NUMBER** for each answer.*

First day at work

- Name of supervisor: **1** .....
- Where to leave coat and bag: use **2** ..... in staffroom
- See Tiffany in HR: to give **3** ..... number
  to collect **4** .....
- Location of HR office: on **5** ..... floor
- Supervisor's mobile number: **6** .....

Questions 7–10

Complete the table below.

*Write **ONE WORD ONLY** for each answer.*

Responsibilities			
	Task 1	**Task 2**	**Notes**
Bakery section	Check sell-by dates	Change price labels	Use **7** ..... labels
Sushi takeaway counter	Re-stock with **8** ..... boxes if needed	Wipe preparation area and clean the sink	Do not clean any knives
Meat and fish counters	Clean the serving area, including the weighing scales	Collect **9** ..... for the fish from the cold-room	Must wear special **10** .....

PART 2 *Questions 11–20*

Questions 11 and 12

*Choose **TWO** letters, **A–E**.*

Which **TWO** problems with some training programmes for new runners does Liz mention?

A There is a risk of serious injury.
B They are unsuitable for certain age groups.
C They are unsuitable for people with health issues.
D It is difficult to stay motivated.
E There is a lack of individual support.

Questions 13 and 14

*Choose **TWO** letters, **A–E**.*

Which **TWO** tips does Liz recommend for new runners?

A doing two runs a week
B running in the evening
C going on runs with a friend
D listening to music during runs
E running very slowly

Questions 15–18

What reason prevented each of the following members of the Compton Park Runners Club from joining until recently?

*Write the correct letter, **A**, **B**, or **C** next to Questions 15–18.*

Reasons	
A	a lack of confidence
B	a dislike of running
C	a lack of time

Club members

15 Ceri .....

16 James .....

17 Leo .....

18 Mark .....

Questions 19 and 20

*Choose the correct letter, **A**, **B** or **C**.*

19 What does Liz say about running her first marathon?

A It had always been her ambition.
B Her husband persuaded her to do it.
C She nearly gave up before the end.

20 Liz says new runners should sign up for a race

A every six months.
B within a few weeks of taking up running.
C after completing several practice runs.

PART 3 *Questions 21–30*

Questions 21–25

*Choose the correct letter, **A**, **B** or **C**.*

21 Kieran thinks the packing advice given by Jane's grandfather is

A common sense.
B hard to follow.
C over-protective.

22 How does Jane feel about the books her grandfather has given her?

A They are not worth keeping.
B They should go to a collector.
C They have sentimental value for her.

23 Jane and Kieran agree that hardback books should be

A put out on display.
B given as gifts to visitors.
C more attractively designed.

24 While talking about taking a book from a shelf, Jane

A describes the mistakes other people make doing it.
B reflects on a significant childhood experience.
C explains why some books are easier to remove than others.

25 What do Jane and Kieran suggest about new books?

A Their parents liked buying them as presents.
B They would like to buy more of them.
C Not everyone can afford them.